

Review

The path ahead for understanding Toll-like receptor-driven systemic autoimmunity

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Five mammalian Toll-like receptors (TLR 3, 7, 8, 9, and 13) recognize nucleic acids (NA) and induce signals that control the function of multiple immune cell types and initiate both innate and adaptive immune responses. While these receptors enable recognition of diverse microbial threats, in some instances, they respond inappropriately to self-NA released from host cells and drive the development of autoimmune diseases. Specifically, activation of TLR7 and TLR8 by self-RNA and TLR9 by self-DNA has been linked to development of a collection of systemic autoimmune or autoinflammatory disorders, including systemic lupus erythematosus, systemic juvenile idiopathic arthritis, and macrophage activation syndrome. Here, we discuss recent progress in understanding how these receptors contribute to such diverse clinical phenotypes. We highlight how comparative studies between mice and humans have not only been beneficial in identifying key pathways relevant for disease but also reveal gaps in our understanding of disease mechanisms. We identify several challenges that we hope the field will tackle in the years ahead.

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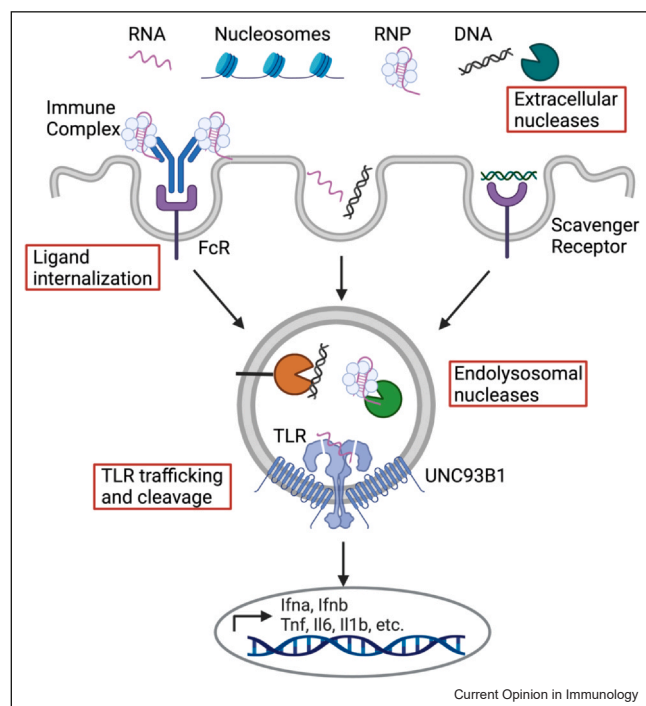
Introduction

Toll-like receptors (TLRs) are a family of transmembrane innate immune sensors that localize either on the

cell surface or within the endolysosomal system and recognize diverse microbial ligands, including nucleic acids (NA) by TLRs 3, 7, 8, 9, and 13. The ability of TLRs to discriminate between microbial and self-NA is mediated by cell biological mechanisms that limit receptor access to self-NA ligands rather than strict molecular discrimination [1] (Figure 1). All NA-sensing TLRs localize to endosomes and lysosomes, and multiple mechanisms co-operate to confine receptor activation to these intracellular compartments. The trafficking chaperone UNC93B1 is critical for the localization and function of NA-sensing TLRs. This multipass transmembrane protein binds NA-sensing TLRs in the endoplasmic reticulum, facilitates their transport to endolysosomes, and can also regulate TLR activation and turnover within endolysosomes [2–4]. Activation of surface localized TLRs by extracellular NA is limited because the ectodomains of NA-sensing TLRs must undergo cleavage by endolysosomal proteases before gaining functional competency [1]. Additionally, extracellular and endolysosomal nucleases degrade NA before they trigger TLRs [1]. Finally, the level of TLR7 and TLR8 in endosomes is carefully regulated such that small increases can enable responses to self-NA [1]. Thus, discrimination between microbial and self-NA is achieved through compartmentalization of receptor activation and careful tuning of receptor and ligand levels. While the specific mechanisms regulating these aspects for each NA-sensing TLR can differ, the overall effect is the same: that NA ligands reaching endosomal TLRs are principally microbial in origin.

Genetic evidence from both humans and mice has established that recognition of self-NA by TLR7, TLR8, and TLR9 can result in autoimmune diseases, which involve both innate immune responses and antigen-specific adaptive immune responses, or autoinflammatory diseases, which involve mainly innate cell responses [5]. These findings underscore the importance of the regulatory mechanisms discussed above to protect from these diseases. While increased signaling of each of these endosomal TLRs leads to systemic diseases, the specifics of the diseases caused by each differ perhaps more than would be expected, suggesting fundamental differences between TLR7, TLR8, and TLR9. Also, the role of each receptor may differ in mice and humans, particularly as TLR8 is not functional in mice. Here, we discuss how similarities and differences between TLR7, TLR8, and TLR9 may dictate the specific diseases with

Figure 1



Regulation of TLR7/8/9 responses. Extracellular RNA and DNA or RNA and DNA within protein complexes can be degraded by extracellular nucleases before being internalized by a variety of cell surface receptors or by nonspecific mechanisms. Once within the endosomal system, RNA and DNA are further processed by endolysosomal nucleases, in some cases leading to generation of TLR7/8/9 ligands but in other cases leading to their degradation. TLR7/8/9 bind UNC93B1 in the endoplasmic reticulum and traffic with this chaperone through the Golgi to endolysosomes. UNC93B1, in conjunction with other cellular factors, can influence the level of TLRs in endosomes (not shown) by regulating receptor delivery as well as receptor turnover. Once in endolysosomes, the TLR7/8/9 ectodomain is cleaved, allowing for optimal ligand binding and receptor signaling, which induces downstream signaling and a variety of transcriptional responses. Each of these key steps can be regulated to determine the level and outcome of TLR7/8/9 signaling. RNP, ribonucleoprotein.

which they are associated. We review evidence from mice and humans, and we discuss how consideration of both species will be necessary to achieve a complete understanding of the mechanisms underlying systemic autoimmunity and autoinflammatory diseases driven by endosomal TLRs. Finally, we outline our vision for future studies needed to better understand this topic.

Toll-like receptor 7

Of all the endosomal TLRs, our understanding of TLR7-driven autoimmune disease is the most complete, which can largely be attributed to similarities between mice and humans. It has long been clear in mouse models that TLR7 is a key driver of systemic lupus erythematosus (SLE) through the use of TLR7-deficient mice; however, the role of TLR7 in human SLE had not

been conclusively established until more recently. Children with coding mutations in *TLR7* that enhance responses to self-NA develop SLE and/or neuro-inflammatory disease [6–8], and mice engineered to encode one of these mutations develop severe SLE-like disease [6]. Increased expression of TLR7 in mice, either via insertion of bacterial artificial chromosomes (BAC) containing the *Tlr7* locus or duplication of the *Tlr7* gene in the Yaa (Y-linked autoimmune accelerator) locus in which a portion of the X chromosome, including *Tlr7* is translocated onto the Y chromosome, also promotes SLE-like disease [9,10]. Recent work shows that incomplete X chromosome inactivation in female immune cells results in expression of TLR7 from both chromosomes, suggesting that TLR7 copy number may also contribute to the female sex bias in SLE [11,12]. Analyses of mice and humans with mutations in *UNC93B1* show that alterations in TLR7 trafficking and/or turnover result in SLE [13–19].

The strong association between TLR7 activation by self-NA and SLE is likely due to the expression of TLR7 in both B cells and plasmacytoid dendritic cells (pDCs) in humans and mice. Antinuclear antibodies (ANA) and other autoantibodies contribute to organ damage in SLE, and B cell-intrinsic TLR7 signaling is required for ANA production and expansion of auto-reactive B cells, both in germinal center and extra-follicular responses [6,9,20–22]. Internalization of self NA-containing antigens via B cell receptor (BCR) crosslinking delivers self-NA to endosomal TLR7 in NA-specific B cells, bypassing the compartmentalization that normally limits such recognition (Figure 1). TLR7 signaling in pDCs leads to potent production of type I interferon (IFNs), key cytokines in SLE pathogenesis that enhance immune responses in many ways, including by upregulating nucleic acid sensors, including TLR7. pDC responses to self-NA are also facilitated by ANA, as pDCs can internalize self NA-containing immune complexes via Fc receptors (FcRs). A defining feature of SLE is a type I IFN signature in the blood [23,24]. IFN α receptor (IFNAR) deficiency reduces many TLR7-driven disease hallmarks in mice [25–27], and IFNAR blocking antibodies are now approved for the treatment of SLE [28]. Thus, expression of TLR7 enables activation of B cells and pDCs by self-NA, and each cell type reinforces activation of the other. Such feed-forward mechanisms likely explain why TLR7 dysregulation is so consequential for the development of SLE.

In mouse models, overexpression of TLR7 also causes macrophage activation syndrome (MAS) or secondary hemophagocytic lymphohistiocytosis (sHLH) [29]. sHLH is a potentially fatal hyperinflammatory cytokine storm syndrome that is called MAS when it develops secondary to rheumatologic diseases, as opposed to infection or malignancy. In humans, MAS occurs most

commonly secondary to systemic juvenile idiopathic arthritis (sJIA); however, it also occurs secondary to SLE and other rheumatic diseases. The inciting factor of MAS is unclear, but given the similarity to primary HLH, in which defects in cytolytic pathways in T and NK cells lead to IFN γ -dependent macrophage activation during chronic viral infection, similar pathways may be involved [30]. In TLR7-overexpressing mice, MAS is characterized by myeloid expansion, anemia and thrombocytopenia and the TLR7-dependent differentiation of specialized monocyte-derived cells that phagocytose red blood cells called inflammatory hemophagocytes, upon which disease depends [29]. Therefore, while TLR7 signaling in B cells and pDCs likely drives the underlying SLE in TLR7-overexpressing mice, TLR7 signaling in monocytes most likely causes MAS in this model. Chronic topical administration of the TLR7 agonist R848 also induces MAS-like disease in mice [31], so the development of MAS is not just a quirk of TLR7 overexpression by artificial means.

Toll-like receptor 8

The role of TLR8 in both infection and autoimmune/ autoinflammatory diseases is less understood than that of TLR7, principally because a functional RNA-sensing TLR8 protein is not expressed in mice. TLR7 and TLR8 share the highest sequence similarity among all TLR family members and arose through gene duplication. Nevertheless, human TLR8 has distinct RNA-binding preferences from human TLR7, and TLR8 expression is limited to neutrophils, monocytes, macrophages, and some DCs, which also distinguishes it from TLR7. The lack of expression of TLR8 on pDCs and B cells predicts that, unlike TLR7, TLR8 activation will not drive type I IFN-mediated autoimmune diseases, such as SLE, and evidence from both humans and mice supports this prediction.

Alleles of *TLR8* with enhanced sensitivity to RNA ligands have been identified recently [32,33]. Unlike SLE and neurological diseases in individuals with TLR7 gain-of-function (GOF), a range of inflammatory phenotypes occur with TLR8 GOF alleles. Common manifestations include severe neutropenia and other cytopenias, low IgG, hepatomegaly and splenomegaly, and increased pro-inflammatory cytokines; however, autoimmune or autoinflammatory diseases were also present in some individuals, including autoimmune thrombocytopenia, type 1 diabetes, and hemophagocytosis, similar to that seen in MAS. High expression of TLR8 from a BAC transgene in mice caused early lethality with severe multiorgan inflammation with circulating ANA and pro-inflammatory cytokines. At low transgene expression, mice did not develop spontaneous disease or ANA but were more susceptible to collagen-induced arthritis [34]. In humans, TLR8 has been linked

to the autoinflammatory diseases sJIA and adult-onset Still's disease, though whether self-NA or viral-NA are involved is not clear [34,35]. Together these mouse and human data show that TLR8 GOF associates with autoinflammatory diseases, consistent with the principal expression on monocytes, macrophages, and neutrophils. Notwithstanding, the presence of autoantibodies in several of the TLR8 GOF patients highlights the continuum between autoinflammation and autoimmunity.

There are additional complexities beyond the lack of a functional TLR8 in mice that make disentangling the functions of TLR7 and TLR8 difficult. Though TLR8 and TLR7 have distinct synthetic ligands, viral or self-RNAs such as those found in ribonucleoproteins are complex and typically can activate both receptors. Additionally, there have been reports that TLR7 and TLR8 cross-regulate each other's signaling when both are simultaneously activated [36], that the expression of TLR7 and TLR8 mRNA are reciprocally regulated [34,37], and that TLR7 and TLR8 activate distinct signaling cascades in cells that express both receptors, such as monocytes and neutrophils [36,38]. Therefore, the outcome of TLR8 signaling may depend not only on the RNA ligand, but on the relative expression of TLR7 in a given cell. For example, TLR7 signaling in CD14⁺ monocytes induces a weak signal, while TLR8 signaling drives a potent transcriptional response that dominates when a dual TLR7/TLR8 ligand such as R848 is used [36,38]. These examples underscore how little we understand of TLR8 regulation, both in isolation and in conjunction with TLR7.

Toll-like receptor 9

Of all the TLRs with the potential for self-NA recognition, evaluating the contribution of TLR9 toward disease is perhaps the most difficult. Unlike TLR7 and TLR8, no patients with GOF mutations in *TLR9* have been described. Without such a 'smoking gun', the field has relied on mouse models for evidence that TLR9 responses to self DNA are pathological. However, the expression of TLR9 differs between mouse and human, making it difficult to directly correlate disease phenotypes in mice with disease in humans. While human and mouse B cells and pDCs both express TLR9, mouse monocytes, but not human monocytes, are highly TLR9 responsive. Moreover, studies in mice have provided conflicting evidence regarding the role of TLR9 activation in disease.

The most direct evidence that TLR9 activation alone can drive disease comes from mice subjected to repeated TLR9 ligand injections and from two genetically engineered mouse strains. Mice injected chronically with CpG oligodeoxynucleotide develop MAS-like disease characterized by anemia, splenomegaly, and hepatitis, which depends on IFN γ [39,40]. Similar disease occurs

in mice lacking PLD4, an endolysosomal exonuclease that degrades DNA and RNA [41]. The pathology in PLD4^{-/-} mice depends on TLR9 and IFN γ [41], as does disease in mice expressing a mutant version of TLR9 that is better activated by self-DNA [42]. Why these TLR9-driven diseases resemble MAS and require IFN γ while TLR7-driven disease is more SLE-like and dependent on type I IFN is unclear, especially considering that both TLR7 and TLR9 are expressed by pDCs and B cells, two key pathogenic cell types for SLE.

In other mouse models that involve TLR9 activation, disease can be more complex because of contributions from other NA sensors that impact pathogenesis. This is true for mice lacking DNase1L3, which primarily functions to degrade extracellular cell-free DNA, particularly that associated with microparticles released by dying cells. DNase1L3-deficient mice develop an IFNAR-dependent SLE-like disease, but deficiency of both TLR9 and TLR7 is required to rescue disease [43]. A similar TLR7 and TLR9 dependence is seen in mice lacking both the PLD3 and PLD4 exonucleases, which die at 2–3 weeks of age from MAS-like disease. This disease is only partially rescued by TLR9 or IFN γ deficiency because the absence of both endolysosomal nucleases also leads to activation of TLR7 and cGAS/STING (cyclic GMP-AMP synthase/stimulator of interferon response cGAMP interactor), likely because PLD3 and PLD4 have activity on RNA as well as DNA substrates [44,45]. Finally, PLD4-deficient mice develop a more SLE-like disease on the Balb/c genetic background than the B6 background, which remains TLR9 dependent, indicating that modifier genes can influence how disease manifests [46].

In contrast to TLR9 activation promoting disease, in several SLE mouse models, the absence of TLR9 results in worse disease. This observation was first made in MRL/lpr mice, where TLR9 deficiency exacerbates disease, while TLR7 deficiency rescues disease and has subsequently been shown in other systems [21,47]. This protective role for TLR9 has been extended to several additional mouse models of SLE, but the mechanism of protection remains unclear. Recent studies have ruled out that TLR9-deficiency increases TLR7 protein levels or TLR7 signaling [48], but this remains an area of active investigation. An intriguing study tracking the development of B cells in humanized mice found that the absence of TLR9 promoted the development of polyreactive B cells [49]. Whether increased frequencies of such cells underlie the more severe disease in certain TLR9-deficient mouse models has not been established, and a link between reduced TLR9 responses and SLE in humans has not been found to date. Additionally, some mouse models of SLE show a partial TLR9 dependence, suggesting a pathogenic role of TLR9,

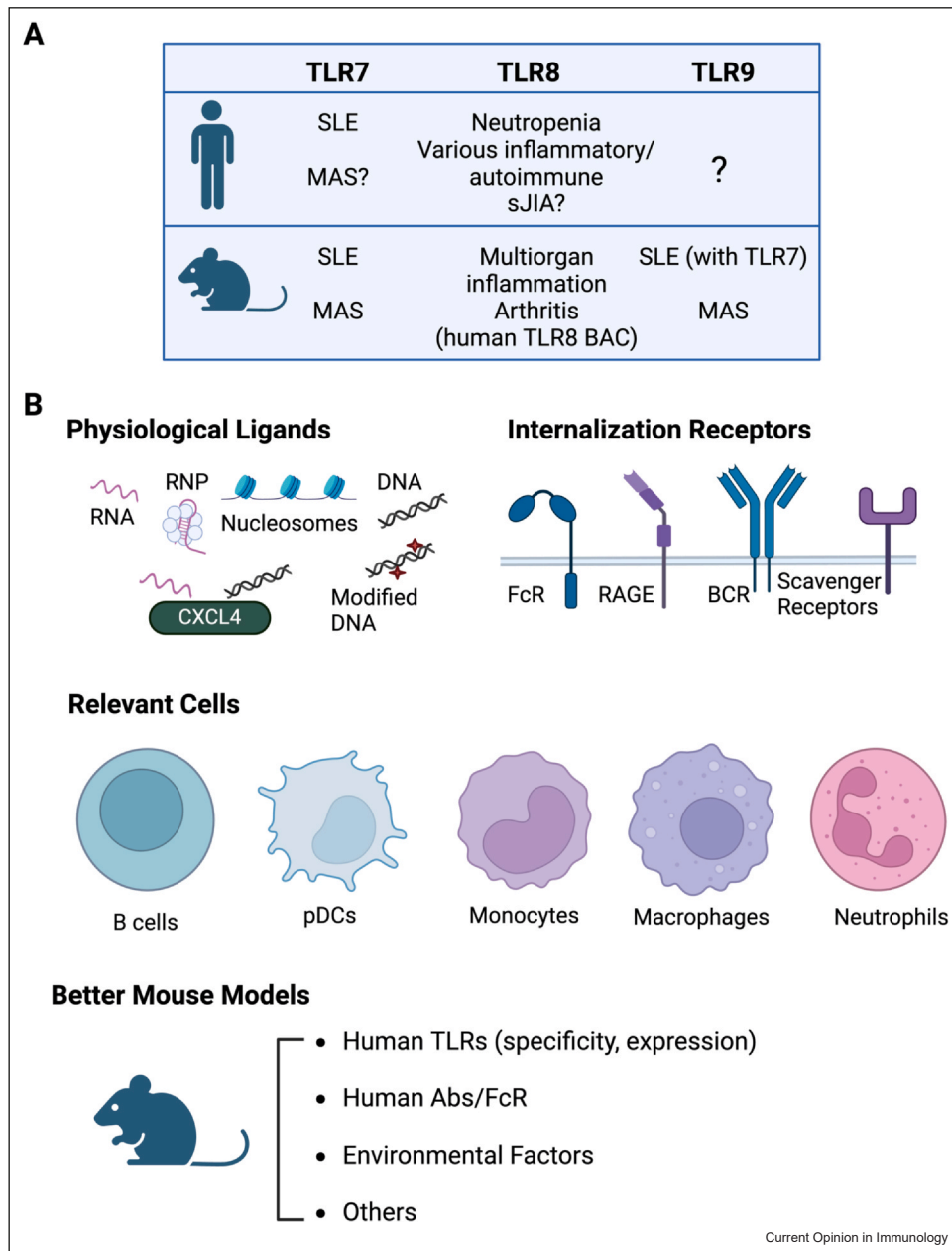
including in PLD4-, DNase1L3-, and TNIP1-deficient mice, which is not consistent with a protective role for this TLR [43,46,50].

The path forward

In the ~25 years since the seminal discovery that TLR recognition of self-NA-containing immune complexes can drive activation of rheumatoid factor-specific B cells [51], the concept that self-NA recognition by TLRs can drive systemic autoimmunity and/or inflammation has become an established paradigm. Studies of both mice and humans have strengthened this paradigm, as have basic studies of endosomal TLR function in immune cells. As discussed here, based on our current understanding of TLR biology we are beginning to understand the diverse clinical presentations associated with TLR7/8/9-driven disease (Figure 2a), but significant gaps in our understanding remain. We believe that filling these gaps will require broader studies of endosomal TLR function and regulation in both human cells as well as improved mouse models that better recapitulate aspects of TLR function specific to humans (Figure 2b).

While it is now well accepted that TLR7 is instrumental in driving both human and mouse SLE, we still lack a clear understanding of the range of clinical phenotypes that each of the NA-sensing TLRs can drive in humans and of the determinants of those phenotypes. For example, why does TLR7 induce MAS in mice but less so in humans, or is this aspect of TLR7-driven disease simply not yet well-recognized in humans? To that end, individuals with TLR7-driven SLE have cytopenias that may hint of MAS, and it has recently been recognized that MAS is more common in childhood SLE than previously thought [52]. Alternatively, the lower responsiveness of human monocytes to TLR7 ligands may not drive MAS-like disease as efficiently as in mice [38], although the expression of TLR8 in these cells makes any definitive assessment of the importance of TLR7 difficult. Individuals with TLR8 GOF alleles have varied clinical phenotypes, and our understanding of the cell types and signaling networks driving disease in such individuals remains limited. We also have little understanding of the role of TLR9 in human autoimmune and autoinflammatory diseases, in large part due to the lack of individuals identified with *TLR9* GOF alleles. Experimentally generated *Unc93b1* mutations that selectively increase TLR9 signaling have been identified [15]; however, such TLR9-skewing variants have yet to be identified in humans, unlike for *UNC93B1* alleles that favor TLR7 signaling. For all of the NA-sensing TLRs, developing a better understanding of the key cell types in which each receptor functions and to what extent those functions are conserved between mice and humans is a critical next step for the field.

Figure 2



The path forward in understanding TLR7/8/9 in systemic autoimmune/ autoinflammatory diseases. **(a)** Our current understanding of the contribution of TLR7/8/9 to systemic diseases in mouse and human. **(b)** Four areas where future studies should focus to better understand this topic — use of physiological ligands, identifying the receptors used for internalization of these ligands and how they contribute to responses, studying responses in all relevant cell types, and building better mouse models to recapitulate human disease for mechanistic studies. RNP, ribonucleoprotein; Abs, antibodies.

The use of small nucleoside analogs (R848, Imiquimod, TL8-508, etc) or synthetic oligos with reduced degradation (CpG-A/B/C, ssRNA40, etc.) has been instrumental in our understanding of NA-sensing TLRs, but to better understand the activation and regulation of these receptors in the context of disease, we should use more physiological ligands and methods of internalization. There is a growing list of endogenous RNA and

DNA ligands capable of activating TLRs, including endogenous retroelements and retroviruses, ribonucleoproteins, mitochondrial DNA, and nucleosomes. NA can also be modified, which can increase or decrease their ability to activate TLRs. The mechanisms and kinetics of internalization will be distinct for these complex and varied ligands compared with synthetic ligands, as will their processing within the endolysosomal system.

Binding of NA to self-proteins may facilitate internalization, trafficking, and TLR recognition, as was originally shown for the cationic peptide LL37 that binds DNA and promotes TLR9 responses [53]. More recently, the chemokine CXCL4 has been shown to serve a similar function for both RNA and DNA promoting recognition by TLR8 and TLR9, respectively [54–56]; however, in B cells, CXCL4 has been proposed to sequester DNA away from TLR9-containing endosomes [49]. This is a new and exciting area of investigation with important implications for autoimmunity, as CXCL4 is highly upregulated in disease. Additionally, NA, particularly DNA but perhaps also RNA, may be modified during cell death or cell stress pathways before release, including during NETosis, pyroptosis, and apoptosis, and these sources of ligands have not been well studied as TLR7/8/9 ligands. The method of internalization of these physiological ligands may also affect kinetics of ligand processing and TLR activation. Most *in vitro* studies use nucleoside ligands that are taken up passively into cells or use transfection reagents to deliver DNA or RNA oligos. However, in systemic autoimmune diseases such as SLE, autoantibody-antigen complexes are internalized via FcRs, modified nucleic acids may be internalized by receptors such as receptor for advanced glycation endproducts (RAGE), and the B cell receptor mediates internalization of ligands in B cells. Studying these relevant autoantigen ligands may unearth disease-specific mechanisms of regulation that have not been elucidated with synthetic ligands.

As a field, we have made huge strides in understanding how TLRs function in homeostasis and disease using mouse models that recapitulate important aspects of human disease. As we learn more about human phenotypes associated with GOF TLR7/8/9 signaling, any path forward to understand molecular pathogenesis of TLR-driven autoimmune and autoinflammatory diseases will require new mouse models and the power of mouse genetics to complement studies of clinical phenotypes and *ex vivo* human cells. Incorporation of multiple, independent mouse models can help to minimize any bias associated with a particular model. We must also invest in fully defining the biology of TLR7/8/9 signaling in humans, across diverse cell types and within the heterogeneity of these cell types, including monocytes, tissue macrophages, DCs, and B cells, as the vast majority of current studies have focused on macrophages (mouse) or macrophage cell lines (human). An important problem going forward is how to make mouse models that more faithfully recapitulate the TLR7/8/9 and cell-type specific biology of humans. One solution is to use humanized mice with both strong myeloid and lymphoid reconstitution such as the MISTRG mice [57]. However, it is not yet clear to what extent they recapitulate human-specific aspects of TLR-driven autoimmune or autoinflammatory diseases. Alternatively, one might attempt to humanize

the mouse immune system in ways that are most directly relevant for TLR-driven disease, for example by ‘fixing’ mouse TLR8 to make it responsive to ligands. However, to do so, we need to understand fundamental differences between mouse and human TLR7/8/9 biology. For example, mouse monocytes strongly respond to TLR9 agonists, whereas human monocytes do not. There may also be important differences not only in the receptors themselves but also in proteins that regulate aspects of their signaling. For example, we recently identified Fc α R/CD89 as a key receptor along with Fc γ RIIA on pDCs for internalizing human SLE immune complexes; however, neither Fc α R nor the IgA1 heavy chain exists in the mouse genome, and mouse pDCs express a distinct IgG receptor [58]. Thus, to engineer this human pathway into mice would be complex. We acknowledge that fully modeling the human immune system in mice with the goal of understanding TLR7/8/9 in autoimmune and autoinflammatory disease will always be a challenge. However, as a field, we should let the perfect be the enemy of the good.

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Data Availability

No data were used for the research described in the article.

Declaration of Competing Interest

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